

Book Recommendations at Scale: A Hybrid Neural Approach on the Goodreads Dataset

Stanford CS229 Project

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Abstract

We present a hybrid neural recommendation system for books, trained on the Goodreads dataset comprising 15.7 million reviews across 2.36 million books and 465,000 users. Our pipeline progresses from a sentiment-based global Top-N recommender to a personalized Neural Collaborative Filtering (NCF) model, and finally to a hybrid neural network that combines pretrained collaborative filtering embeddings with author embeddings and 20-dimensional content features — enabling per-user recommendations tailored to individual reading histories. We apply two-phase transfer learning to stabilize training and incorporate an element-wise interaction term inspired by He et al. (2017). Models are evaluated using RMSE on held-out validation and temporal test sets, and qualitatively assessed using an LLM-as-a-Judge framework. Our best collaborative filtering model achieves a validation RMSE of 1.0875, and our best hybrid model achieves 1.0900, demonstrating that dense interaction data largely subsumes content signals. However, qualitative evaluation via LLM-as-a-Judge reveals a nuanced picture: hybrid recommendations significantly outperform CF for niche readers (+0.312 relevance delta, $p < 0.001$) while CF performs better for mainstream readers (−0.140, $p = 0.013$), demonstrating that content features provide strongest signal precisely where collaborative filtering is weakest.

1 Introduction

With an estimated 129 million unique books published since the invention of the Gutenberg press (Taycher, 2010), readers face substantial challenges in discovering titles that align with their interests. Unlike movie or music recommendation, book consumption is time-intensive — the cost of a poor recommendation is high. This information overload, combined with the limited time of modern readers, underscores the need for intelligent recommendation systems that effectively leverage large-scale user-generated content. An effective system must balance personalization — leveraging what a user has liked before — with discovery of new books they might enjoy.

The input to our system is a user’s historical book reviews — each containing a rating (1–5 scale), review text, user ID, and book ID — along with book metadata including genres, language, publication year, and author information. We use neural networks operating on user ratings, book metadata, and VADER-extracted review sentiment to output a predicted rating $\hat{r}_{ui} \in \mathbb{R}$ for any (user, book) pair, which is then used to rank unread books and generate a personalized Top-N recommendation list.

We make the following contributions:

- A production-quality end-to-end recommendation pipeline processing the *full* Goodreads dataset — all 15.7M reviews, 2.36M books, and 465K users — without sub-sampling, progressing from sentiment-based global ranking to personalized recommendations.

- Scalable data engineering across the full dataset — including Parquet-based columnar storage for fast loading, lazy chunk processing to fit within memory constraints, and pre-computed lookup tensors for efficient inference.
- A hybrid neural architecture combining pretrained NCF embeddings with author embeddings and 20-dimensional content features, trained using two-phase transfer learning and an element-wise user–book interaction term.
- An empirical ablation study over 7 model iterations that analyzed overfitting, temporal drift, and the marginal value of content features on dense datasets.
- An LLM-as-a-Judge evaluation framework with user segmentation analysis complementing RMSE, revealing statistically significant differences in recommendation quality across niche and mainstream reader segments.
- A custom PyTorch implementation without relying on off-the-shelf recommendation libraries.

2 Related Work

Collaborative Filtering. Matrix factorization (Koren et al., 2009) decomposes user-item interactions into latent factors; He et al. (2017) extended this with NCF, replacing dot products with an MLP to capture non-linear interactions.

Content-Based Filtering. Content-based methods (Pazzani and Billsus, 2007) recommend items based on feature similarity, effective for cold-start but limited in capturing latent preferences.

Hybrid Models. Zhang et al. (2019) show content features add most value in sparse regimes; Covington et al. (2016) demonstrate hybrid models outperform pure CF at YouTube scale.

Book Recommendation. Wan and McAuley (2018) analyze the Goodreads dataset specifically, noting the richness of shelf and genre metadata.

3 Dataset and Features

Dataset. We use the UCSD Goodreads dataset (Wan et al., 2019), comprising 15.7 million reviews, 2.36 million books, and 465,000 users. Each review contains a user ID, book ID, rating (1–5), and a `read_at` timestamp recording when the user finished the book — used later for temporally ordered train/validation/test splitting. Book metadata includes title, authors, genres, language code, and publication year.

Pre-processing. While all 15.7M reviews and 2.36M books are processed for sentiment analysis and global Top-N ranking, we filter to users with at least 10 ratings and books with at least 25 ratings for personalized recommendations, yielding 9,044,974 reviews across 190,502 users on 89,424 books. The lower book threshold (25 vs. the typical 50) is justified by our hybrid model’s ability to supplement sparse collaborative filtering signal with content features. Global Top-N recommendations retain the stricter threshold of 50 reviews for statistical confidence in sentiment scores.

Train/Validation/Test Split. We apply a per-user time-based split using the `read_at` timestamp: for each user, their oldest 70% of reviews form the training set, the next 15% the validation set, and the most recent 15% the test set. Every user appears in all three splits, eliminating user cold-start. This temporal ordering simulates real deployment conditions and explains the observed gap between validation RMSE (earlier time period) and test RMSE (later time period) due to temporal preference drift.

Content Features. We engineer 20 content features per book:

- **Genre features (10):** Shelf counts across 10 genre categories (fiction, fantasy/paranormal, history/biography, romance, mystery/thriller, non-fiction, children, comics/graphic, poetry, young-adult), expressed as a fraction of total shelf placements per book.

- **Language features (6):** Language represented as a one-hot encoding across 6 groups — English, Spanish, German, French, Portuguese, and Other — where all remaining languages are mapped to the Other category.
- **Numerical features (4):** Publication year, average sentiment score (VADER compound, normalized from $[-1, 1]$ to $[0, 1]$), mean rating, and review count — all min-max normalized using training set statistics only, and applied consistently across train, validation, and test sets to prevent data leakage.

4 Methods

4.1 Global Top-N (Sentiment Baseline)

Our simplest recommender ranks books by their vote-weighted average VADER (Hutto and Gilbert, 2014) sentiment score, filtered to books with at least 50 reviews. This non-personalized baseline surfaces universally well-reviewed books and provides a ceiling for what a user-agnostic system can achieve.

4.2 Neural Collaborative Filtering

Following He et al. (2017), our NCF model learns dense embedding representations for users and books:

$$\hat{r}_{ui} = f(\mathbf{e}_u, \mathbf{e}_i; \Theta) \quad (1)$$

where $\mathbf{e}_u \in \mathbb{R}^{64}$ and $\mathbf{e}_i \in \mathbb{R}^{64}$ are user and book embeddings, and f is a multi-layer perceptron with architecture $[128 \rightarrow 64 \rightarrow 1]$ with ReLU activations and dropout. Embeddings are initialized with $\mathcal{N}(0, 0.01)$ to prevent large initial predictions; linear layers use Kaiming Uniform initialization (He et al., 2015) appropriate for ReLU networks.

We train with Adam (Kingma and Ba, 2014) ($\text{lr} = 0.001$, batch size 2048) — chosen for its adaptive per-parameter learning rates which handle sparse embedding gradients effectively — with MSE loss, dropout $p = 0.35$, early stopping (patience=7), and ReduceLROnPlateau scheduler (patience=3, factor=0.5) that reloads best weights upon learning rate reduction. These hyperparameters were arrived at iteratively — starting from a baseline configuration and incrementally adding the scheduler and tuning dropout — as detailed in the ablation study in Section 5.2.

4.3 Hybrid Neural Network

Our hybrid model extends NCF with two additional input signals:

Author Embeddings. Each book’s primary authors (role=“”) are encoded as a 64-dimensional embedding $\mathbf{e}_a \in \mathbb{R}^{64}$. For books with multiple primary authors (8.8% of books), we average their embeddings. Author IDs are shifted by 1 (reserving index 0 for padding) enabling clean batch processing of variable-length author lists via masked averaging:

$$\mathbf{e}_a = \frac{\sum_k \mathbf{e}_{a_k} \cdot \mathbb{I}[a_k \neq 0]}{\sum_k \mathbb{I}[a_k \neq 0]} \quad (2)$$

Content Features. The 20-dimensional content feature vector $\mathbf{c} \in \mathbb{R}^{20}$ described in Section 3 is concatenated directly.

Interaction Term. Following He et al. (2017), we include the element-wise product of user and book embeddings as an explicit interaction signal:

$$\mathbf{z}_{ui} = \mathbf{e}_u \odot \mathbf{e}_i \in \mathbb{R}^{64} \quad (3)$$

This directly captures per-dimension alignment between user preferences and book characteristics, facilitating learning without requiring the MLP to discover these interactions implicitly.

The full input to the MLP is:

$$\mathbf{x} = [\mathbf{e}_u \| \mathbf{e}_i \| \mathbf{z}_{ui} \| \mathbf{e}_a \| \mathbf{c}] \in \mathbb{R}^{276} \quad (4)$$

with MLP architecture $[276 \rightarrow 128 \rightarrow 64 \rightarrow 1]$.

Two-Phase Transfer Learning. We initialize user and book embeddings from the pretrained NCF model. Naively training all parameters jointly caused the randomly initialized MLP and author embedding layers to produce noisy gradients that disturbed the pretrained representations — evidenced by best validation performance at epoch 1 with degradation thereafter. We address this with two-phase training:

- **Phase 1 (5 epochs):** Freeze user and book embeddings. Train only author embeddings and MLP layers until they stabilize.
- **Phase 2 (up to 50 epochs):** Unfreeze all parameters. Fine-tune jointly at reduced learning rate ($lr = 0.0001$) to allow pretrained embeddings to adapt gently.

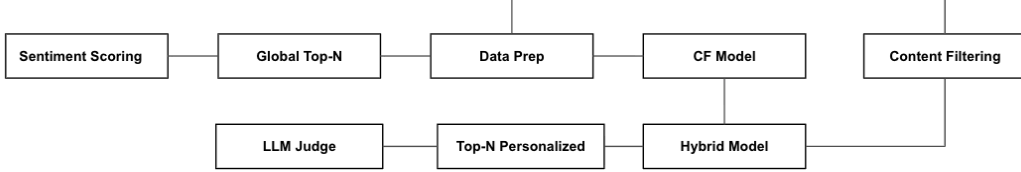


Figure 1: End-to-end recommendation pipeline.

5 Experiments / Results / Discussion

5.1 Evaluation Metric

We evaluate using Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{(u,i) \in \mathcal{T}} (\hat{r}_{ui} - r_{ui})^2} \quad (5)$$

We report validation RMSE (used for model selection) and test RMSE (final evaluation).

5.2 Ablation Study

Table 1 presents our full ablation across 7 model iterations.

Table 1: Ablation study results. Train RMSE reported at best validation epoch.

Model / Configuration	Train RMSE	Val RMSE	Test RMSE
CF — no scheduler, dropout=0.20	0.9655	1.0919	1.2867
CF — scheduler, dropout=0.20	0.9792	1.0902	1.2847
CF — scheduler, dropout=0.35	0.9803	1.0875	1.2803
Hybrid — single phase	1.1137	1.0894	1.2830
Hybrid — two phase	0.9948	1.0911	1.2833
Hybrid — two phase + interaction	1.0317	1.0900	1.2810
Hybrid — two phase + interaction + author init	0.9944	1.0930	1.2853

Collaborative Filtering Improvements. Iterative regularization improvements yielded consistent gains: adding the ReduceLROnPlateau scheduler improved validation RMSE from 1.0919 to 1.0902, and increasing dropout from 0.20 to 0.35 further improved it to 1.0875. The scheduler reloads best weights upon learning rate reduction, preventing degradation after the learning rate step.

Hybrid Model Analysis. The hybrid model did not surpass CF on overall RMSE. With 9M interactions, collaborative filtering embeddings likely already capture author and genre signals through correlated rating patterns, causing books by similar authors or genres to cluster in embedding space. This aligns with Zhang et al. (2019), who note that content features add the most value in sparse regimes. However, CF and Hybrid top-10 lists overlap by only 4.87 books on average across 1,000 users, indicating the hybrid model surfaces meaningfully different recommendations despite similar RMSE — a distinction further explored through the segmented LLM-as-a-Judge section 5.3.

Overfitting Analysis. The train–validation gap in the CF model (~ 0.11) indicates mild overfitting, while the larger validation–test gap (~ 0.19) is primarily attributable to temporal preference drift rather than model overfitting. Dropout and early stopping effectively mitigate this training overfit. The same pattern is observed for the hybrid model.

Two-Phase Training. Two-phase training successfully addressed gradient interference from randomly initialized layers, though it did not improve final validation RMSE over single-phase training.

5.3 LLM-as-a-Judge Evaluation

To complement RMSE with qualitative assessment, we use an LLM-as-a-Judge framework (Zheng et al., 2023). We evaluate the top 1,000 most active users (≥ 100 ratings, rating std ≥ 1.25). For each user, we generate Top-10 recommendations from CF and Hybrid models and ask Claude to score relevance (1–10) given the user’s top-10 and bottom-10 rated books, enforcing strict separation so that the highest bottom-10 rating is lower than the lowest top-10 rating.

Table 2: LLM-as-a-Judge relevance scores (1–10 scale, 10 being most relevant) across 1,000 users.

Model	Mean Relevance (1–10)
CF (best: scheduler, dropout=0.35)	3.325
Hybrid (best: two phase + interaction)	3.447

The hybrid model outperforms CF by +0.122 in mean relevance across 1,000 users. Absolute scores are low, but the relative difference between models remains the primary signal of interest. To examine where hybrid recommendations add the most value, we segment users by the average review count of books they rated — a proxy for how mainstream or niche their reading tastes are. Users in the bottom quartile (niche readers, $n = 250$) tend to rate obscure books with sparse collaborative filtering signal, while users in the top quartile (mainstream readers, $n = 250$) rate popular, well-reviewed books.

We assess statistical significance using a paired t-test with $t = \bar{d}/(s_d/\sqrt{n})$, where $\bar{d}$ is the mean per-user relevance delta (hybrid – CF), s_d is its standard deviation, and n is the group size, under the null hypothesis $H_0 : \bar{d} = 0$.

Table 3: LLM-as-a-Judge relevance scores segmented by user reading taste.

User Read Segment	N	CF Mean	Hybrid Mean	Relevance Delta	t-stat	p-value
Niche (bottom quartile)	250	2.668	2.980	+0.312	5.565	< 0.001
Mainstream (top quartile)	250	3.748	3.608	−0.140	−2.507	0.013
Overall	1,000	3.325	3.447	+0.122	4.103	< 0.001

As shown in Table 3, the hybrid model outperforms CF substantially for niche users (+0.312 relevance delta, $t = 5.565$, $p < 0.001$) while CF slightly outperforms hybrid for mainstream users (−0.140, $t = -2.507$, $p = 0.013$). Both findings are statistically significant via paired t-test, confirming that content features provide the strongest signal precisely where collaborative filtering is weakest — sparse, obscure book interactions.

6 Conclusion

We present a full recommendation pipeline progressing from sentiment-based global ranking to a hybrid neural network trained on 9M Goodreads reviews. Our best CF model achieves validation RMSE of 1.0875 (test RMSE 1.2803) and our best hybrid model achieves validation RMSE of 1.0900 (test RMSE 1.2810). Segmented LLM-as-a-Judge evaluation with paired t-tests confirms that content features provide strongest signal where collaborative filtering is weakest — hybrid outperforms CF for niche readers (+0.312, $p < 0.001$) while CF wins for mainstream readers (−0.140, $p = 0.013$). Future work includes cold-start applications and user history embeddings.

7 Contributions

All components — project topic selection, data processing, temporal train/validation/test splitting, sentiment analysis, collaborative filtering, content feature engineering, hybrid neural network architecture, two-phase transfer learning, LLM-as-a-Judge evaluation, statistical significance testing, segmentation analysis, and reporting — were designed and implemented by the author individually on a personal iMac without leveraging cloud compute or external GPUs.

The author used Claude (Anthropic) as an AI collaborator for research exploration, debugging, and report refinement — consistent with the CS229 policy on generative AI use. All design decisions, architectural choices, experimental results, and conclusions are the author’s own.

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